
KAPATE OS

Enterprise Business Operating System — Operations & Production Deployment Manual

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Platform: Kapate OS Unified Business Suite

Organization: Kapate Consultancy

Classification: Confidential / Internal Operational Guide

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Lead Architecture: Kapate Technical Architecture & Systems Engineering

EXECUTIVE NOTICE: This manual contains operational procedures, system access matrices, deployment guides, and disaster recovery runbooks for Kapate OS. Ensure that all deployment credentials and JWT signing keys are managed securely in production environments.

1. Executive Summary & System Architecture

Kapate OS is the proprietary enterprise operating platform designed for **Kapate Consultancy** to centralize, automate, and scale technology consulting operations. The platform integrates lead acquisition, CRM pipelines, milestone-based project management, employee/intern workforce allocation, automated invoice billing, and an isolated client portal into a unified, secure system.

1.1 Architecture Breakdown

- **Presentation Layer (Port 3001):** Next.js 16 (React 19, TypeScript, Tailwind CSS, Lucide Icons, Recharts Analytics) configured with an executive white theme.
- **Public Website (Port 3000):** High-performance HTML5/CSS3 marketing portal with seamless redirection into Kapate OS.
- **Application Engine (Port 8000):** FastAPI REST engine running asynchronous Python 3.11+ workers with Pydantic v2 schemas and JWT token validation.
- **Data Storage Layer:** SQLAlchemy 2.0 ORM with SQLite in Write-Ahead Logging (WAL) mode or PostgreSQL 15+ with complete foreign-key cascading and immutable audit logs.

Component	Technology	Port / Binding	Role / Function
Kapate OS Frontend	Next.js 16 / React 19	localhost:3001	Interactive enterprise dashboard & modules
Main Website	Vanilla HTML / CSS / JS	localhost:3000	Public client acquisition & consulting showcase
Core Backend API	FastAPI / Python 3.11	localhost:8000	Business logic, RBAC, JWT auth, database ORM
Database	SQLite (WAL) / PostgreSQL	File / Port 5432	Relational storage & audit event streams

2. Role-Based Access Control (RBAC) & Security Policy

Kapate OS enforces strict Zero-Trust role authorization. Access permissions are verified at both the API endpoint middleware and frontend UI navigation levels.

Role	CRM	Projects	HR / Interns	Finance	Client Portal
SUPER ADMIN	Full	Full	Full	Full	Full Control
PROJECT MANAGER	View/Edit	Full	View/Assign	View Budgets	View Sprints
EMPLOYEE	Assigned	Task Work	Self Clock-in	Self Claims	Restricted

INTERN	No Access	Tasks Only	Training Hub	No Access	No Access
FINANCE CONTROLLER	Deals View	Budgets	Payroll View	Full Invoicing	No Access
CLIENT	No Access	No Access	No Access	Own Invoices	Dedicated View

DATA ISOLATION GUARANTEE: Under no circumstances are employee salaries, company gross margins, internal labor hourly costs, or private HR notes exposed to Client accounts. The Client Portal uses an isolated query serializer.

3. Step-by-Step Module Operations Guide

3.1 CRM & Sales Pipeline

- **Lead Creation:** In the CRM module, click **Add Lead** to log client information, project scope, budget estimation, and lead source.
- **Pipeline Movement:** Drag or update deals across stages: *New Lead -> Discovery Call -> Proposal Sent -> Negotiation -> Contract Won*.
- **Qualification Scoring:** System assigns lead scores (1-100) based on budget alignment, technical feasibility, and timeline.

3.2 Proposals, Contracts & Digital Signatures

- **Generate Proposal:** Convert a qualified deal to an executive proposal detailing deliverables, technology stack, and commercial milestones.
- **E-Signature & Contracts:** Once accepted, issue a binding digital contract with audit timestamps and stakeholder signatory tracking.

3.3 Project Delivery, Sprints & Kanban Board

- **Project Initiation:** Create a dedicated project workspace with budget, target delivery date, Project Manager, and allocated team.
- **Kanban Tasks:** Manage sprint tasks across *TODO, IN PROGRESS, IN REVIEW, CHANGES REQUESTED, COMPLETED*.
- **Milestone Sign-Off:** Complete deliverables to trigger milestone approvals and invoice billing.

3.4 Workforce & Dedicated Intern Hub

- **Employee Management:** Track skill matrices, active project utilization percentages, and engineering capacity.
- **Intern Mentorship:** Mentors evaluate interns across 4 core competencies (Technical execution, Problem-solving, Communication, Milestone delivery).
- **Certificate Generator:** Produce official verifiable **Certificate of Completion** PDF documents upon internship conclusion.

3.5 Time Tracking, Attendance & Leaves

- **Daily Attendance:** Employees and interns log clock-in and clock-out timestamps with location tags.
- **Timesheets:** Log billable and non-billable hours per project task for manager sign-off.
- **Leave Requests:** Manage sick leave, casual leave, and compensatory off balances with managerial approvals.

3.6 Financial Invoicing & Profitability Audits

- **Milestone Invoicing:** Automatically generate GST-compliant invoices upon milestone completion.
- **Payment Reconciliation:** Log received client bank transfers, transaction IDs, and outstanding balances.

- **Profitability Audit:** Real-time analytics monitor billed revenue vs. actual developer payroll costs and server infrastructure expenses.

3.7 Isolated Client Portal

- **Client Transparency:** External stakeholders access a clean, restricted portal showing project milestones, completed tasks, and invoice download links.

4. Production Deployment & Server Setup Guide

4.1 Server Requirements & Prerequisites

- **Operating System:** Ubuntu 22.04 LTS (64-bit) or Windows Server 2022
- **Compute Resources:** 2 vCPUs, 4 GB RAM, 40 GB NVMe Storage minimum
- **Runtimes:** Node.js 20+ LTS, Python 3.11+, Git, Nginx
- **Process Managers:** Systemd (Backend FastAPI daemon) and PM2 (Next.js frontend daemon)

4.2 Backend Production Deployment (FastAPI)

```
# 1. Setup Python Virtual Environment
cd /var/www/kapate-consultancy/backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt

# 2. Configure Systemd Service (/etc/systemd/system/kapate-backend.service)
[Unit]
Description=Kapate OS FastAPI Enterprise Backend
After=network.target

[Service]
User=www-data
Group=www-data
WorkingDirectory=/var/www/kapate-consultancy/backend
ExecStart=/var/www/kapate-consultancy/backend/venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8000 --workers 4
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
```

4.3 Frontend Production Deployment (Next.js)

```
# 1. Build Standalone Production Next.js Bundle
cd /var/www/kapate-consultancy/kapate-os-frontend
pnpm install --frozen-lockfile
pnpm build

# 2. Daemonize with PM2 Process Manager
pm2 start npm --name "kapate-os-frontend" -- start -- -p 3001
pm2 save
pm2 startup
```

4.4 Nginx Reverse Proxy & SSL Configuration

```
server {
    listen 443 ssl http2;
    server_name os.kapateconsultancy.com;

    ssl_certificate /etc/letsencrypt/live/os.kapateconsultancy.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/os.kapateconsultancy.com/privkey.pem;

    location / {
        proxy_pass http://127.0.0.1:3001;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection 'upgrade';
        proxy_set_header Host $host;
    }

    location /api/ {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
    }
}
```

5. Database Maintenance, Backups & Runbook

5.1 Automated Daily Database Backups

```
# Daily cron backup job at 02:00 AM
0 2 * * * cp /var/www/kapate-consultancy/kapate_os.db /var/backups/kapate-os/kapate_os_$(date +%Y%m\%d_%\H%M%S).db
# Delete backups older than 30 days
0 3 * * * find /var/backups/kapate-os/ -type f -name "*.db" -mtime +30 -delete
```

5.2 Operational Health Checks & Troubleshooting

Issue / Symptom	Diagnostic Step	Resolution Procedure
API 502 / Offline	curl http://127.0.0.1:8000/api/v1/health	Restart backend: sudo systemctl restart kapate-backend
Port 3001 in use	lsof -i :3001 or netstat -ano	pm2 restart kapate-os-frontend or kill stale process
Database Locked	Check SQLite journal mode	Enable WAL: PRAGMA journal_mode=WAL;
Auth Token Expired	Inspect JWT header exp	User re-authenticates or calls refresh token

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